METRIC

MIL-R-51015C(EA)
27 September 1990
SUPERSEDING
MIL-R-51015B(EA)
4 September 1987

MILITARY SPECIFICATION

RIOT CONTROL AGENT, CS CAPSULE (METRIC)

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of capsule filled with Riot Control Agent, CS.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1365

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve Test
U-C-115 - Capsule, Gelatin

MILITARY

MIL-R-51029 - Riot Control Agent, CS

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

SPECIAL PACKAGING INSTRUCTION (SPI)

PMIL-R-51015 - Riot Control Agent, CS Capsule

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Capsule. The capsule shall conform to size 00 of U-C-115.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

3.1.2 Riot Control Agent, CS. The Riot (Control Agent, CS used to fill the capsule shall conform to MIL-R-51029 and shall be ground so that no less than 100 percent by weight passes through an 840-micrometer (No. 20) sieve when tested as specified in 4.2.4.1.

3.2 Weight. The filled capsule shall weigh no less than 0.58 gram (g) when tested as specified in 4.2.4.2.

3.3 Workmanship. The filled capsule shall be free from interior or exterior contamination such as dirt, grease, and oil. The filled capsule shall be free from damage such as cracks, dents, and punctures The exterior of the capsule shall not be contaminated with Riot Control Agent, CS.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.3.) When

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the filled capsules offered for acceptance at one time which have been produced by one manufacturer, at one plant, during one 8-hour shift, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of Riot Control Agent, CS shall be used in one lot of filled capsules.

4.2.2 Sampling.

TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with table I. Samples shall be selected at random.

4.2.2.2 For destructive tests. Fifty filled capsules shall be randomly selected from each lot.

4.2.3 Inspection procedure. Sample items and the packaging thereof shall be examined and tested as specified in 4.2.3.2 and 4.2.3.3. Failure of any sample item or packaging thereof to conform to any characteristic in the classification of characteristics or failure of the Riot Control Agent, CS in the filled capsules to meet the particle size requirement shall be cause for rejection of the lot represented.

4.2.3.1 Inspection characteristics. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.2.3.2 For examination and nondestructive tests. Sample filled capsules and the packaging thereof shall be examined in accordance with table I and the following classification of characteristics:

(a) Riot Control Agent, CS Capsule.

Category	Characteristic	Sampling and acceptance criteria	Inspection method
Critical:	None defined		
Major			
101	Assembly	Table I, level V	Visual
102	No damage	Table I, level V	Visual
103	No interior contamination	Table I, level V	Visual
104	Weight	Table I, level V	4.2.4.2
Minor			
201	No exterior contamination	Table I, level VI	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

(b) Packaging (SPI PMIL-R-51015).

Category	Characteristic	Sampling and acceptance criteria	Inspection method
Critical:	None defined		
Major			
101	Unit pack container leakage	Table I, level IX	4.2.4.3
102	Item completely clean and dry prior to unit packing	Table I, level VI	Visual
103	Contents per container evident and correct	Table I, level VI	Visual
104	Unit pack container and closure evident and correct	Table I, level VI	Visual
105	Exterior coating on cans evident and correct	Table I, level Vl	Visual
106	Side seam of can striped	Table I, level VI	Visual
107	Tape closure for bottle evident and correct	Table I, level VI	Visual
108	Packing container and closure evident correct	Table I, level VI	Visual
109	Packing assembly evident and correct	Table I, level VI	Visual
110	Marking evident, correct, and legible	Table I, level VI	Visual

4.2.3.3 For destructive tests. The Riot Control Agent, CS in the filled capsules taken in 4.2.2.2 shall be tested as specified in 4.2.4.1.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Particle size. Weigh to the nearest 0.01 g approximately 8 g of the Riot Control Agent, CS and place into an 840-micrometer (No. 20) sieve conforming to RR-S-366 nested into a bottom pan. Cover the sieve and shake by hand for 3 minutes. Weigh the material in the bottom pan to the nearest 0.01 g. The difference between the initial and final weighings shall not exceed 0.03 g.

4.2.4.2 Weight. Weigh the filled capsule on a balance which has an accuracy of no less than 0.001 g.

4.2.4.3 Unit pack container leakage. The unit pack container (can) containing the Riot Control Agent, CS shall not leak when tested in accordance with method 5009.2 of FED. TEST METHOD STD. NO. 101 using the hot water technique.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI PMIL-R-51015.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Intended use. Riot Control Agent, CS Capsules are intended for use in the training of troops.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)
(c) Level of packing required

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Safety information on CS. For safety information on CS, see Hazardous Component Safety Data Statement (HCSDS) No. 20067. Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7210.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-Army Quadripartite Standardization Agreement 427. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Particle size
Weight

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Custodian:

Army - EA

Preparing activity:

Army – EA

Project No. 1365-A100

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Chemical Research, Development
and Engineering Center
ATTN: SMCCR-PET-S
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:47Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-51015C(EA)	RIOT CONTROL AGENT, CS CAPSULE (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION FOR NEW DESIGN

METRIC

MIL-R-51015C(EA)
NOTICE 1
16 August 1995

MILITARY SPECIFICATION

RIOT CONTROL AGENT, CS CAPSULE (METRIC)

This notice should be filed in front of MIL-R-51015C(EA) dated 27 September 1990.

MIL-R-51015C(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity

Army - EA

Project No. 1365-A107

AMSC N/A

FSC 1365

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:47Z
Check the source to verify that this is the current version before use.